SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID –** [fill in — I'll provide]
- **Problem Statement Title –** Real-time weather data collection, misinformation triage, and analysis platform for disaster response
- **Theme –** Disaster Management
- **PS Category –** Software
- **Team ID –** [fill in]
- **Team Name (Registered on portal) –** [fill in]

Team: [fill in]

IDEA TITLE: National Weather Big Data Analytics Platform + WeatherGPT

Proposed Solution & System Capabilities (Working Prototype):

- **Multi-Source Ingestion:** Ingests 4 weather APIs (Open-Meteo, OpenWeather, Tomorrow.io, WeatherAPI), citizen reports, and news/social feeds (GDELT, Reddit) into a single pipeline.
- **AI Classification & Triage:** NLP engine classifies reports into 13 hazard categories with calibrated confidence scores [0.45–0.98].
- **Multimodal Deduplication:** Collapses duplicates using Haversine distance (≤3.5km) + temporal delta (≤90m) + Jaccard token overlap without discarding velocity metrics.
- **Explainable Trust Scoring:** Evaluates credibility [0–100] across 6 transparent physical signals (radar consistency, spatial bounds, temporal freshness, cross-corroboration, media).
- **PostGIS Incident Clustering:** Groups reports into spatial hazard incidents (≤12km, ≤4h), computes a 0–100 Platform Impact Score, and escalates high-velocity hotspots to Early Warning feeds.
- **Grounded WeatherGPT:** Provides emergency teams a grounded, citation-backed conversational interface — zero hallucinated figures, strictly factual live telemetry.
- **Core Innovation:** Multi-provider consensus scoring + transparent 6-signal trust audit (not a black-box filter) + human-in-the-loop verification console with permanent audit trails.

Team: [fill in]

TECHNICAL APPROACH & ARCHITECTURE

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Technologies & Tools Employed:

- **Backend & Queue:** FastAPI (Python async ASGI), Celery worker pool + Redis 7 for asynchronous background ingestion & reclustering.
- **Database & GIS:** PostgreSQL 15 + PostGIS 3.4 for geodesic spatial indexing (ST_DWithin, SRID 4326), GeoAlchemy2, ACID audit history.
- **Frontend UX:** Next.js 14 App Router, React 18, TypeScript, Tailwind CSS (dark command palette), Leaflet GIS, Recharts telemetry.
- **AI / ML Pipeline:** Calibrated 13-category lexical NLP classifier (<1.5ms latency), Haversine + Jaccard deduplication, OpenCLIP zero-shot support.
- **WeatherGPT Reasoning:** Google Gemini 1.5 Flash primary → OpenAI GPT-4o-mini secondary → Rule-based deterministic fallback with citation evidence panels.

Methodology & Processing Flow (10-Stage Pipeline):

01. Weather Feeds
4 APIs + Citizens

02. Normalization
UTC, GPS, Schema

03. AI Classifier
13 Hazard Classes

04. Deduplication
Haversine ≤ 3.5km

05. Trust Engine
6-Signal Scoring

06. PostGIS Store
Spatiotemporal Clust

07. Impact Score
0–100 Severity Index

08. Early Warning
Hotspot Escalation

09. Admin Console
Human Verification

10. WeatherGPT
Grounded Decision AI

Team:
[fill in]

FEASIBILITY AND VIABILITY ANALYSIS

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Feasibility & Operational Readiness:

- **Operational Prototype Built & Validated:** Fully implemented codebase with 62/62 automated pytest suites passing across providers, NLP classifier, dedup, clustering, and WeatherGPT.
- **Low-Latency Ingestion & UI:** Sub-15ms end-to-end report ingestion pipeline; Next.js 14 production build compiled with 15 static routes and 0 errors.
- **100% Offline Demonstration Capability:** Pre-cached meteorological dataset and deterministic 500-report generator (demo/seed.py) ensure zero failure risk from external connectivity during hackathon judging.
- **Economical & Horizontally Scalable:** Operates on free-tier global models and open-source PostGIS/Redis stack; stateless ASGI architecture enables linear horizontal scale.

Key Challenges & Architectural Mitigations:

- **Challenge 1 (No Public IMD Radar API):** Mitigated via 4-provider global model consensus (Open-Meteo, OWM, Tomorrow, WeatherAPI) cross-validated with local citizen reports and official IMD linkouts.
- **Challenge 2 (Spam & Rumor Vulnerability):** Mitigated via automated 6-factor credibility scoring, rate-limiting, and mandatory human review before unverified posts become ground truth.
- **Challenge 3 (Regional Language Nuances):** Mitigated with hybrid bilingual lexicon (English/Hindi idioms like 'aandhi', 'loo', 'cloudburst') and human review queue for low-confidence text.

Team:
[fill in]

IMPACT AND BENEFITS TO STAKEHOLDERS

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Value Proposition & Societal Returns:

- **Target Beneficiaries & Users:** National & State Disaster Management Authorities (NDMA/SDMA), district collectors, Emergency Operations Centers (EOCs), municipal flood control cells, and NDRF/SDRF field response units.
- **Life-Saving Social Impact:** Accelerates triage during flash floods, cloudbursts, and cyclones from hours to seconds; enables timely targeted citizen evacuations and localized shelter management.
- **Economic & Resource Efficiency:** Dramatically reduces wasteful dispatch of emergency personnel and pumps triggered by viral social media rumors, stale photos, or duplicate panic reports.
- **Unified Operational Picture:** Bridges the dangerous gap between macro satellite forecasts and chaotic ground eyewitness posts into a single verified, geospatial situational dashboard.
- **Accountability & Governance:** Human-in-the-loop verification station creates an immutable audit trail for every emergency declaration, ensuring institutional transparency and post-event analysis.

Team:
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RESEARCH, REFERENCES AND STANDARDS

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Meteorological APIs & Spatial Platforms:

- **Open-Meteo Weather API:** High-resolution global weather models (ECMWF, GFS), hourly forecasts, free tier — https://open-meteo.com/
- **OpenWeatherMap API:** Current atmospheric conditions, wind gusts, atmospheric pressure API v2.5 — https://openweathermap.org/api
- **Tomorrow.io Intelligence API:** 1-hour hyper-local precipitation and convective storm tracking API v4 — https://www.tomorrow.io/
- **WeatherAPI.com:** Real-time observational feed and condition code cross-validation — https://www.weatherapi.com/
- **PostGIS Spatial Database:** Geodesic geometry indexing (ST_DWithin, WGS-84 SRID 4326) — https://postgis.net/

National Disaster Guidelines & Global Standards:

- **GDELT 2.0 Global News Project:** Real-time news intelligence monitoring for disaster keywords — https://www.gdeltproject.org/
- **NDMA Incident Response System:** National Disaster Management Guidelines on Incident Command & Response, Govt. of India (2010)
- **WMO Multi-Hazard Guidelines:** World Meteorological Organization Guidelines on Multi-Hazard Early Warning Systems (MHEWS-II)
- **USGS Earthquake & Hazard Feed:** Real-time GeoJSON seismic event protocols & spatial schemas — https://earthquake.usgs.gov/